

Before the Machine

From quarks to the physical requirements of fusion

We will build the machine from its purpose. First we must understand the matter we intend to change, the energy available from that change, and the obstacle between possibility and a useful reaction rate.

0.1. Inside a nucleon

A quark is one of the elementary constituents in our present theory of matter. To locate it, begin with an atom: a positively charged nucleus surrounded by negatively charged "quantum clouds" of electrons. Electrons are also elementary constituents of matter and carry charge $-e$. Chemical reactions change electron arrangements while leaving the nuclei intact. Nuclear reactions such as fission and fusion change the nuclear arrangement itself.

Inside the nucleus are *nucleons*: Protons and neutrons are baryons— composite particles made of three quarks. Quarks come in different flavors. Two types of quarks are up quarks and down quarks. A proton can be written as uud and a neutron can be written as udd , where u and d denote up and down quarks. Quarks possess both electric charge and color charge.

The electromagnetic force is described by rotations of a circle, called the $U(1)$ group, which transforms via a matrix representation of the groups linear map $\chi(\theta) = \exp(iq\theta)$ that satisfies a circles compactification: $\chi(\theta + 2\pi) = \chi(\theta)$, which quantizes charge into an integer, $q \in \mathbb{Z}$. Physics doesn't tell us why the electric charge is $+2e/3$ for an up quark and $-e/3$ for a down quark, experiments do. Given this fact, because constituent phase factors multiply (ie $e^{iq_1\theta}e^{iq_2\theta} = e^{i(q_1+q_2)\theta}$), protons have $+e$ electric charge, and neutrons have *zero net electric charge*:

$$q_p = \left(\frac{2}{3} + \frac{2}{3} - \frac{1}{3}\right) e = e, \quad q_n = \left(\frac{2}{3} - \frac{1}{3} - \frac{1}{3}\right) e = 0.$$

These are *quark labels*: they only specify net quark content. A nucleon is a quantum state that also contains gluon fields and quark–antiquark fluctuations.^[1]

What binds that state? Quarks second kind of charge, called *color*. The theory of color interactions is *quantum chromodynamics* (QCD), and it describes the strong nuclear force— one of the four fundamental forces of nature. Gluons mediate the strong force and possess color charge, leading to self-interactions amongst gluons. This is unlike photons, neutrally charged electromagnetic force carriers that have no direct electric-charge self-interaction, unlike gluons with color charge.

Color is not a single number q to be summed into a different state of color charge — it is a basis vector of $\mathbb{C}^3$, such as $(1, 0, 0)$, to be rotated via matrices whose inverse equal it's conjugate transpose and whose determinant equals to 1, eg elements of $SU(3)$. A state is *color neutral* when it is left invariant by every such rotation: for baryons this is achieved by carrying exactly one quark of each color, combined in the unique totally antisymmetric combination, $\epsilon_{ijk}q^iq^jq^k$, that $SU(3)$ leaves unchanged. This explains why three quarks can form a color-neutral state, whereas two or four quarks alone cannot: three colors demand three quarks to form this invariant, fully analogous to how a determinant vanishes unless you antisymmetrize over as many vectors as there are dimensions.^[2]

Most nucleon mass is not the sum of the small up- and down-quark mass parameters. It comes from the energy of the interacting quark–gluon system. Lattice QCD, which approximates the fields on a spacetime grid and controls extrapolations, has reproduced the light-hadron mass spectrum. Most ordinary matter's mass is in nucleons, made of dynamical QCD energy: typically this makes up over 99% of atomic mass.^[3]

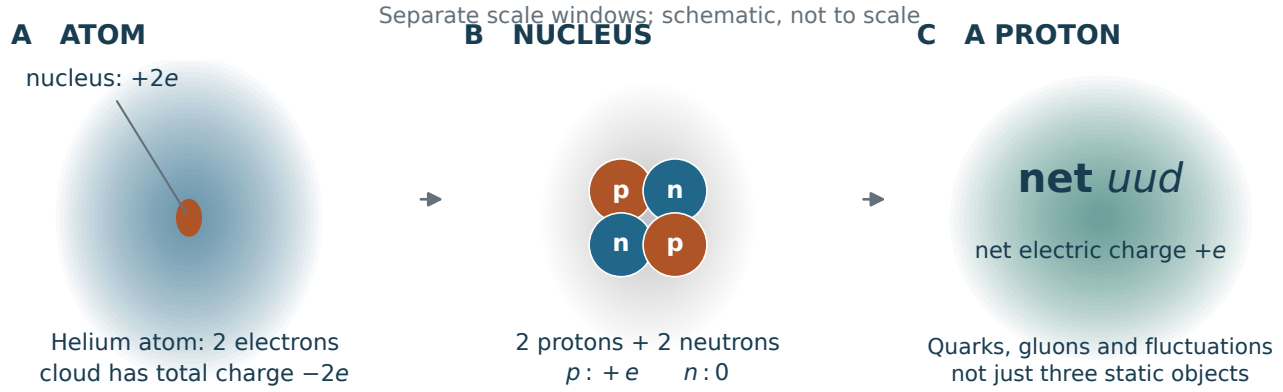


Figure 1: Inside a helium atom is the nucleus, inside the nucleus are the nucleons: protons and neutrons. Inside a nucleon such as a proton are three quarks. Colored regions are schematic; the proton's uud reflect two up-flavored quarks and one down quark. Neutrons follow udd .

0.2. Confinement, and the boundary of the explanation

QCD has an unusual scale dependence. Quarks are never observed isolated in nature. Nucleons are three quarks grouped together, an example of baryons. More generally, any number of quarks grouped together are called mesons.

When quarks are extremely close together, the strong interaction between them becomes weak, and they behave almost like free particles. This is called *asymptotic freedom*. However, as quarks are pulled farther apart, their color field becomes concentrated into a narrow tube of gluon field called a *flux tube*. Stretching this tube requires more and more energy. Eventually, enough energy is stored in the tube that it becomes energetically favorable to create a new quark–antiquark pair precisely when the flux tube breaks. In conclusion, no isolated quark is produced: each separated quark instead joins with one of the newly created partners to form a new color-neutral particle. In this way, quarks are never observed in isolation. This is called *quark confinement* or *color confinement*.^[4]

The flux-tube picture is supported by QCD calculations and lattice simulations, but mathematically proving color confinement analytically from four-dimensional QCD remains an open problem. The key feature is constant linear growth: once the flux tube has formed, each additional unit of separation stores approximately the same additional amount of field energy. This linear confinement seems to produce an *area law*. A simple model keeps the tube's thickness, or cross section, roughly fixed as its length changes. The quark separation (r) therefore parametrizes the constant linear growth of the potential energy.

$$V(r) \simeq V_0 + \mathcal{T}_s r, \quad F_r = -\frac{dV}{dr} \simeq -\mathcal{T}_s.$$

A sharp question now appears. Gluons themselves are massless, yet they account for nearly 99% of regular atomic mass (formed of up and down quarks: heavier quarks can make gluon flux tubes energy mere correction terms). How can massless gluons produce so much mass? By Einstein's equivalence of mass and energy, we know that energy can contribute to mass; the deeper question is what values that mass is allowed to take. Can the energy stored in flux tubes produce a continuous range of masses, from zero up to the maximum energy such tubes can hold before breaking? Or is there instead a smallest positive mass that the tubes can give rise to, some value $\Delta > 0$? This possible empty interval above the vacuum is the *Yang–Mills mass gap*, one of the deepest open problems in mathematical physics.

A Millennium Problem The Yang–Mills existence and mass-gap problem asks for a rigorous construction of the four-dimensional quantum theory and a strictly positive energy gap above its vacuum. Writing H for its energy operator and setting vacuum energy to zero, the gap condition is

$$\text{spec}(H) \cap (0, \Delta) = \emptyset \quad \text{for some } \Delta > 0.$$

Simulations suggest that no physical excitation can have an energy strictly between zero and Δ . Providing a conceptual mathematical proof is expected to shed deep insight into the structure of the universe. Confinement is the physical mechanism that parametrizes energy by the length of flux tubes; the mass gap may be understood as a question about the spectrum $\text{spec}(H)$ of those energy levels. Mathematically, the spectrum is the set of all λ for which the operator $T - \lambda I$ does not have a bounded inverse. This generalizes the concept of eigenvalues from finite-dimensional vector spaces to the infinite-dimensional *Hilbert spaces* found in quantum mechanics.^[6]

Gluons can themselves form bound states called *glueballs*. In 2026, the BESIII experiment in China experimentally confirmed the existence of glueballs via the $X(2370)$ particle, giving experimental substance to the idea that the carriers of a force can organize into matter.^[20]

Color confinement and the Yang Mills mass gap are open questions, and different research directions exist. What most agree on is that the energy in gluon flux tubes is so great that it "leaks out" of nucleons (i.e., protons and neutrons) and forms the foundation of nuclear binding energy, and that the energy leaks out of the nucleus, and this residual energy is responsible for nuclear fusion. There is no consensus for how to bridge hadron formation (flux tubes) with nuclei formation (residual binding energy) with heavy atom formation (nuclear fusion).

A speculative picture. The *liquid-drop model* offers a potential bridge from confinement to nuclear binding to fusion. It pictures the nucleus as a self-bound liquid drop. The drop can deform, vibrate, and respond collectively, but deforming its surface costs energy.^[19] Could an analogous language of collective stiffness help us picture how the quark–gluon field confines its constituents and develops a spectrum of massive excitations?

Imagine two nucleons close together. Their internal fields and fluctuations participate in a shared interaction described through meson exchange and other effective couplings inherited from QCD.^[8] What reaches beyond each nucleon is its capacity to correlate with its neighbours. When the combined state has lower energy than the separated nucleons, the difference is nuclear binding energy. Within the resulting drop, surface vibrations can themselves mediate additional attraction, as developed in *The Nuclear Cooper Pair*.^[19] This suggests a succession of organization: quark–gluon dynamics builds nucleons, interactions among nucleons build nuclei, and collective motion reshapes their binding. Fusion opens a path between nuclear arrangements; when the products are more tightly bound, the energy difference emerges as motion or radiation. For deuterium and tritium, that difference is about 17.6 MeV.^[11] The question inviting further thought is how this sequence produces such different scales: the large rest energy of a nucleon, the smaller binding energy of a nucleus, and the collision energies at which fusion becomes appreciable.

Ordinary liquids suggest a provocative electromagnetic comparison: their cohesion emerges from the charge structure and quantum correlations of their constituents. Rutherford's 1920 conjecture pictured a neutron as a proton tightly combined with an electron; a distinct modern pion-cloud description includes a virtual proton accompanied by a negatively charged pion.^{[21][22]} Both images prompt a question about how internal charge structure can allow a neutral composite to interact with its surroundings. How far could polarization and quantum exchange take an electromagnetic account of nuclear cohesion, and could collective response help us understand the barrier through which fusing nuclei approach?

The *Kondo effect* supplies a further way to think about how electromagnetic phenomenon can mirror nuclear phenomenon such as asymptotic freedom: an antiferromagnetic exchange interaction between a magnetic impurity and a metal's electrons grows strong as the energy scale decreases, producing collective screening below a characteristic Kondo scale.^[7] This also invites us to ask whether similar patterns of scale evolution can illuminate the hierarchy from color confinement to nuclear binding, and how the surrounding medium influences access to a fusion reaction.

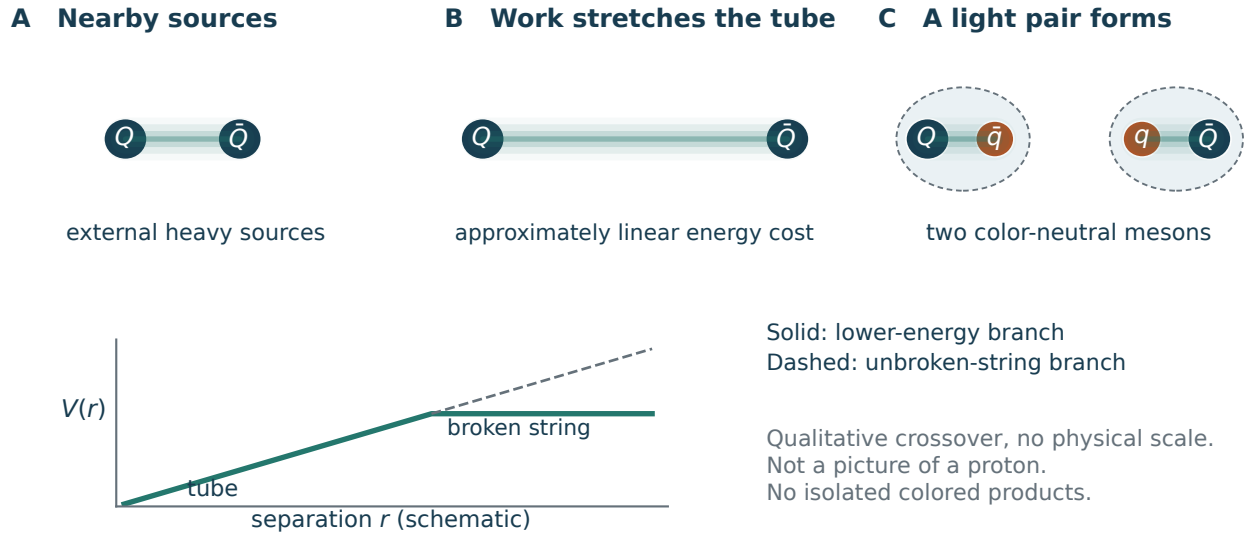


Figure 2: Static heavy sources $Q, \bar{Q}$ and creation of a light $q, \bar{q}$ pair. The final groups are color neutral; dashed ovals group constituents rather than marking physical walls. The qualitative potential sketch omits the smooth mixing near string breaking and has no physical scale. Original schematic based on source 5; not a proton model.

0.3. From confined quarks to interacting nuclei

To connect this picture with a calculation, begin with the interaction between color-neutral nucleons. Their composite structure allows them to exchange energy and momentum while remaining color neutral. The resulting residual interaction provides the link from the internal dynamics of a nucleon to the binding of a nucleus.

At nuclear energies, a practical description replaces explicit quark and gluon variables by nucleons and, in one common formulation, pions. Pions are light quark–antiquark composite particles. Their virtual exchange describes the long-range part of the effective nuclear interaction; shorter-range terms and forces involving three or more nucleons are also needed. These are parts of an effective field theory, organized by QCD's symmetries and a low-energy expansion, with parameters constrained by data. Its accuracy has a regime and an uncertainty.^[8]

Binding is an energy difference between shared states. The energy of interacting nucleons depends on their joint quantum state. A bound arrangement has lower rest energy than the same nucleons separated; reaching it releases the difference into other degrees of freedom, such as the motion of reaction products. This gives the physical content of the bridge from QCD to fusion, which we can now express through measured nuclear masses.

1. Reading the reaction before using it

Let Z count protons and N count neutrons; $A = Z + N$ counts nucleons. The symbol A_ZX specifies a nucleus. Changing Z changes the element X ; changing N at fixed Z gives a different isotope. For our fuel and products,

Nucleus	Symbol	Constituents	Electric charge
Deuterium (deuteron)	$D = {}^2_1\text{H}$	$1p + 1n$	$+e$
Tritium (triton)	$T = {}^3_1\text{H}$	$1p + 2n$	$+e$
Helium-4 (alpha)	$\alpha = {}^4_2\text{He}$	$2p + 2n$	$+2e$
Free neutron	n	unbound neutron	0

The principal D–T fusion channel is

$$D + T \longrightarrow \alpha + n. \quad (1)$$

Its bookkeeping is $2 + 3 = 4 + 1$ nucleons and $1 + 1 = 2 + 0$ units of charge. The neutron on the right is not inside the helium nucleus. Energy, momentum, and angular momentum must also balance, but these conditions do not force an encounter to fuse: the nuclei can scatter without changing species.^[9]

The Sun’s proton–proton route includes $p + p \rightarrow D + e^+ + \nu_e$, where the weak interaction converts a proton into a neutron during formation of deuterium; e^+ is a positron and ν_e an electron neutrino. Our D–T reaction already has neutron-containing fuel and needs no such weak conversion.^[10]

2.1. The energy is a difference between arrangements

We take rest-energy equivalence, $E_{\text{rest}} = mc^2$, as an established physical input, with c the speed of light. Define the positive nuclear binding energy by

$$B(Z, N) = [Zm_p + Nm_n - m(Z, N)]c^2. \quad (2)$$

Here $m(Z, N)$ is the mass of the bare nucleus, not the neutral atom. The definition compares that nucleus at rest with the same numbers of isolated protons and neutrons at rest. Its rearrangement is $m(Z, N)c^2 = Zm_pc^2 + Nm_nc^2 - B(Z, N)$.

The energy made available by one D–T reaction, called its Q value, is the initial rest energy minus the final rest energy. Substituting the three binding energies shows exactly what cancels:

$$\begin{aligned} Q &= (m_D + m_T - m_\alpha - m_n)c^2 \\ &= [(m_p + m_n)c^2 - B_D] + [(m_p + 2m_n)c^2 - B_T] \\ &\quad - [(2m_p + 2m_n)c^2 - B_\alpha] - m_nc^2 \\ &= B_\alpha - B_D - B_T. \end{aligned} \quad (3)$$

The nucleons’ large rest energies cancel. We are not extracting all the QCD energy that constitutes their masses. Evaluated mass data give, rounded to 0.001 MeV, $B_D \simeq 2.225$, $B_T \simeq 8.482$, and $B_\alpha \simeq 28.296$ MeV. Thus

$$Q \simeq 28.296 - 2.225 - 8.482 = 17.589 \text{ MeV} \simeq 17.6 \text{ MeV}.^{[11]}$$

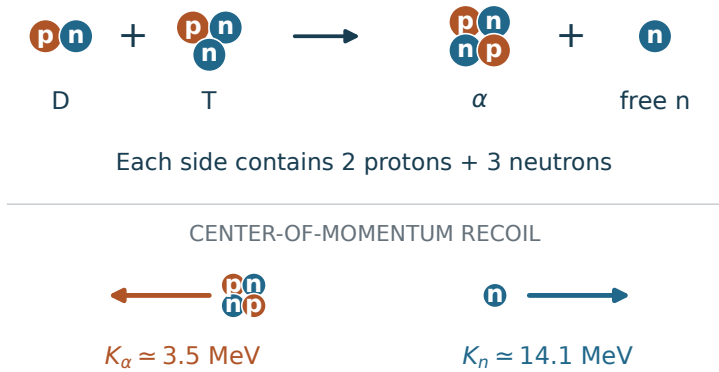
An electronvolt (eV) is the energy gained by charge e across one volt; $1 \text{ MeV} = 10^6 \text{ eV}$ and $1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$.^[12] The rounded binding values are measured inputs, not a QCD calculation performed here.

Where does the released energy go? In the center-of-momentum frame, total momentum is zero. Therefore $\mathbf{p}_\alpha + \mathbf{p}_n = 0$, so the two momentum magnitudes equal some p . For a transparent leading estimate take nonrelativistic kinetic energies and neglect incident kinetic energy compared with Q :

$$K_\alpha = \frac{p^2}{2m_\alpha}, \quad K_n = \frac{p^2}{2m_n}, \quad \frac{K_n}{K_\alpha} = \frac{m_\alpha}{m_n} \simeq 4, \quad K_n + K_\alpha \simeq Q.$$

Consequently $5K_\alpha \simeq Q$, giving $K_\alpha \simeq 3.5 \text{ MeV}$ and $K_n \simeq 14.1 \text{ MeV}$. Precise values use relativistic two-body kinematics and the incident energy; laboratory energies also depend on center-of-momentum motion.^[13]

A Same nucleons, different arrangement



B Rest-energy difference

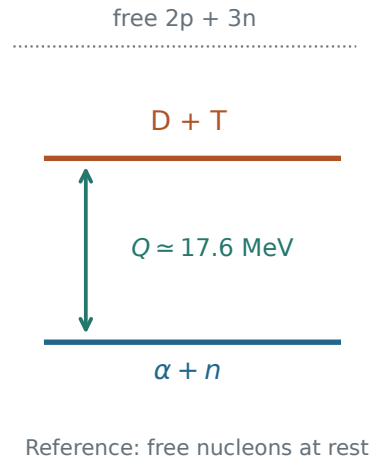


Figure 3: The same $2p+3n$ inventory is rearranged. The two final recoil arrows have equal momentum magnitude, not equal kinetic energy; energies use the leading nonrelativistic estimate in the text. Rest energies are measured relative to separated nucleons. Nucleon circles encode counts, not exact positions or nuclear shapes. Original diagram using sources 11 and 13 and Eq. (3).

2.2. An energetic downhill route can still have a barrier

The wider pattern is visible by plotting binding energy *per nucleon*, B/A , against nucleon count A . It rises broadly from light nuclei to the iron–nickel region and decreases toward very heavy nuclei. Many light-nucleus fusion reactions and heavy-nucleus fission reactions therefore release energy: their products have greater *total* binding energy than their reactants. Neither statement says that every possible fusion or fission channel releases energy, or that matter automatically evolves to iron.^[9]

Why a peak? The effective nuclear attraction is short ranged: a nucleon mostly benefits from nearby neighbours. Small nuclei have a large surface fraction, so many nucleons have fewer neighbours; larger nuclei initially gain more binding per particle. But electric repulsion acts between every pair of protons, making large charged nuclei increasingly costly. This competition gives the broad trend. Quantum shell structure, proton–neutron imbalance, and pairing determine the detailed curve; this argument alone does not calculate the peak.^[9]

Iron is a useful landmark, not a unique universal definition of stability. ^{62}Ni has the largest measured B/A , about 8.795 MeV ; ^{56}Fe is nearby at about 8.790 MeV . Stability against a particular decay still depends on its possible products and conserved quantities.^[11]

Being downhill is not the same as being accessible. A deuteron and triton both carry $+e$. Taking their separation r outside the nuclear-overlap region, Coulomb's law gives a repulsive force $F_r = e^2/(4\pi\epsilon_0 r^2)$. With $U(\infty) = 0$ and $F_r = -dU/dr$, integration gives

$$U(r) = \int_r^\infty \frac{e^2}{4\pi\epsilon_0 s^2} ds = \frac{e^2}{4\pi\epsilon_0 r}. \quad (4)$$

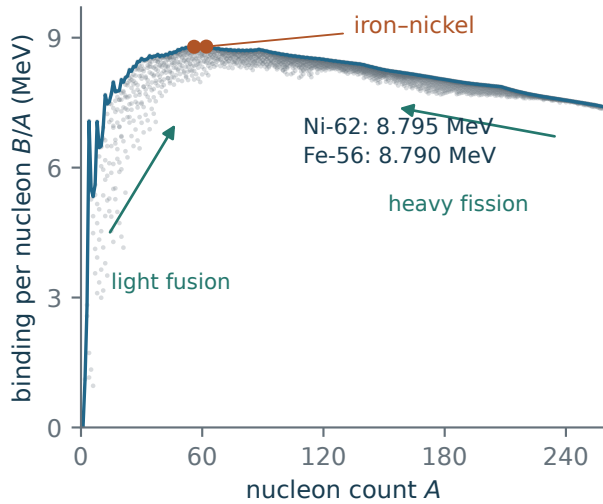
At $r = 5$ fm, where $1 \text{ fm} = 10^{-15} \text{ m}$, this point-charge estimate is $1.440/5 \simeq 0.288 \text{ MeV}$. It illustrates a barrier scale, not a sharp nuclear surface or an exact fusion threshold.

Quantum mechanics permits a wave to penetrate a region forbidden to a classical trajectory. To see why, use a *one-dimensional constant-barrier model* with height $U_0 > E$ and effective mass μ . The stationary Schrödinger equation, an imported quantum law, becomes

$$-\frac{\hbar^2}{2\mu}\psi'' + U_0\psi = E\psi \quad \Rightarrow \quad \psi'' = \chi^2\psi, \quad \chi = \frac{\sqrt{2\mu(U_0 - E)}}{\hbar}.$$

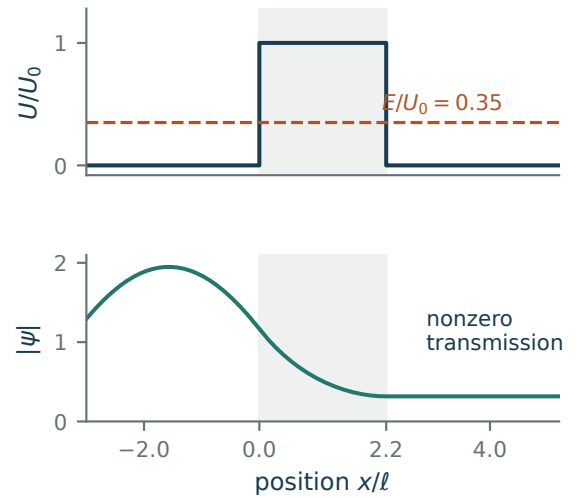
Here ψ is a probability amplitude and $\hbar = h/(2\pi)$ is the reduced Planck constant. The solutions are $Ce^{\chi x} + De^{-\chi x}$, not an identically zero wave. Matching across a finite barrier can leave a transmitted wave. This is tunnelling, not temporary borrowing of energy. The actual D–T problem needs its varying Coulomb field and coupled nuclear channels.^[14]

A Binding energy across nuclei



AME2020 measured entries (no # estimates).
Line: largest tabulated B/A at each A.

B An approach barrier



Toy model: $\ell = \hbar/\sqrt{2\mu U_0}$.
Not a D–T cross-section calculation.

Figure 4: Different horizontal axes, different questions. (A) AME2020 entries with measured binding values and $A \leq 260$; estimated values marked # are excluded. The line is an upper envelope of the discrete entries, not a reaction path. Arrows indicate broad possible energy-releasing trends, not that every channel is favorable. (B) An exact stationary solution of the stated constant-barrier toy model: $E/U_0 = 0.35$, width 2.2ℓ , with $\ell = \hbar/\sqrt{2\mu U_0}$. Matching ψ and its derivative gives nonzero transmission; the lower curve is amplitude magnitude, not an energy curve. Original calculation; sources 11 and 14.

3. From a possible reaction to a probability

The symbol $D+T \rightarrow \alpha+n$ identifies an outcome. To predict its frequency, we need a quantum amplitude. A quantum state can be represented by a vector; time evolution changes that vector according to the interactions. For scattering, an operator S connects specified incoming states to outgoing states. In a simplified orthonormal wave-packet basis, write

$$S|i\rangle = \sum_f a_f |f\rangle, \quad a_f = \langle f|S|i\rangle, \quad P_f = |a_f|^2.$$

$|i\rangle$ denotes the prepared incoming state, $|f\rangle$ a resolved outgoing state, and a_f its complex coefficient. Orthogonality gives $\langle f|g\rangle = 0$ for distinct basis states and 1 when $f = g$; probability conservation then reads $\sum_f |a_f|^2 = 1$. Actual continuum momenta require integrals or finite detector bins rather than a probability for one exact momentum.^[15]

An amplitude includes both magnitude and phase. Contributions to the same indistinguishable outcome add *before* squaring; interference can therefore increase or suppress that outcome. We do not select nuclear products by counting nucleons alone.^[15]

How would we measure this? Send a narrow-energy deuteron beam toward a thin target containing N_T tritons. Let Φ_D be the incident number flux, in $\text{m}^{-2}\text{s}^{-1}$, and let $\dot{N}_{\text{fus}}$ count fusion events per second. In the dilute, independent-target limit, the operational definition is

$$\sigma_{DT}(E) = \frac{\dot{N}_{\text{fus}}}{\Phi_D N_T}. \quad (5)$$

Its units are area. It is the reaction's *effective* area at relative collision energy E , not the literal cross-sectional area of a little hard nucleus. The thin-target assumption avoids appreciable beam depletion and energy loss. Connecting an S -matrix element to this area requires incoming flux normalization, outgoing phase-space integration, and appropriate spin sums or averages; one cannot simply set $\sigma = |a|^2$.^[13]

Barrier penetration is only part of the amplitude. D–T fusion is enhanced by a resonance, a temporarily correlated five-nucleon configuration with the quantum numbers of unbound helium-5. A microscopic calculation by Hupin and colleagues treats incoming $D + T$ and outgoing $\alpha + n$ channels with nuclear interactions from effective field theory. Its high-accuracy comparison includes a small phenomenological resonance adjustment. This gives a concrete bridge from a microscopic nuclear model to measured reaction probabilities, with the model's assumptions still visible.^[16]

Where do path integrals enter? They provide another formulation of amplitudes: sum contributions from alternative histories with phases set by action divided by $\hbar$. Feynman diagrams organize particular expansions; they are not little movies of a measured trajectory. We cite this foundation, but do not claim to evaluate a nonperturbative QCD path integral here.^[17]

4. From one encounter to hot fuel

A reactor needs many reactions, not merely a possible one. Let n_D and n_T denote numbers of deuterons and tritons per unit volume. First imagine all encounters have relative speed v . In a short time dt , one deuteron samples an effective reaction volume $\sigma_{DT}v dt$. That volume contains, on average, $n_T \sigma_{DT}v dt$ tritons. Multiplying by n_D deuterons per unit volume gives the event rate density $\mathcal{R} = n_D n_T \sigma_{DT}v$.

Real particles do not all move at the same speed. Let $g(v) dv$ be the probability that a randomly selected D–T pair has relative speed in $[v, v + dv]$, with $\int_0^\infty g(v) dv = 1$. Averaging the preceding argument gives

$$\langle \sigma v \rangle = \int_0^\infty \sigma_{DT}(v) v g(v) dv, \quad \mathcal{R} = n_D n_T \langle \sigma v \rangle. \quad (6)$$

The brackets mean an average, not multiplication of separately averaged factors. Here $\sigma_{DT}(v)$ is the same cross section, indexed by relative speed rather than collision energy. For unlike species there is no factor 1/2: each D–T pair is counted once. This is a local, dilute, binary-collision model with uncorrelated incoming particles; it is not a formula for an arbitrary strongly correlated state. Dimensional checking gives $[\langle \sigma v \rangle] = \text{m}^3 \text{s}^{-1}$ and $[\mathcal{R}] = \text{m}^{-3} \text{s}^{-1}$. Multiplying by the energy per event gives the local *nuclear* power density, $\mathcal{P}_{\text{fus}} = Q\mathcal{R}$, not net electrical output.

For near-thermal fuel, ion temperature sets the velocity distribution. We keep temperature in kelvin, T_i , distinct from its energy scale, $\Theta_i = k_B T_i$, where k_B is Boltzmann’s constant. For example,

$$\Theta_i = 10 \text{ keV} = 10^4 \text{ eV} \quad \Rightarrow \quad T_i = \frac{10^4}{8.617333262 \times 10^{-5}} \text{ K} \simeq 1.16 \times 10^8 \text{ K}.^{[12]}$$

This is a scale illustration, not a temperature optimum derived from the Coulomb barrier. Evaluated cross sections, energy distributions, and losses must be combined to choose an operating range. The familiar multi-keV D–T regime will be justified by those calculations, not assumed to follow from “hotter is better.”^[18]

In the hot-fuel regime used for magnetic fusion, electrons no longer remain bound to individual nuclei: the fuel is an ionized plasma of positive nuclei and negative electrons. It is not a quark–gluon plasma; nucleons remain confined composite particles.^[1] Positive and negative charge can balance overall even though each constituent responds to electromagnetic fields.

The first machine question How can we give the fuel enough energetic encounters, and retain that energy long enough, for fusion power to compete with losses? In the magnetic-confinement route pursued here, sustained contact between the hot core and material walls would rapidly remove energy and add impurities. We need separation of core fuel from material surfaces. We have not yet earned a torus or a coil: next we ask whether electromagnetic forces can provide the required control.

The charged alpha particle and neutral neutron will later demand different energy-handling strategies. Microscopic color confinement and macroscopic plasma confinement are different problems; the latter does not follow from the former by analogy.

Open sources and the claims they support

These notes make the boundary between derivation and imported physics explicit. All links point to freely accessible material. The body develops its own bookkeeping, energy cancellation, recoil estimate, and rate-counting argument; the underlying physical laws and numerical inputs are not mathematical axioms proved by this thesis.

[1] Constituents and the meaning of quark–gluon plasma.

US Department of Energy, “[DOE Explains: Quarks and Gluons](#).” Introductory orientation for quark content, gluons, and the distinction between intact nucleons and deconfined matter. The electron-cloud description does not assume classical orbital trajectories.

[2] Color charge and gluon self-interactions.

Particle Data Group (2025), “[Quantum Chromodynamics](#),” §9.1, especially Eqs. (9.1)–(9.2) and the discussion of three- and four-gluon vertices on p. 2. These specify the microscopic theory; we do not derive its Lagrangian or Feynman rules in the opening.

[3] Nucleon mass from dynamical QCD.

S. Dürr et al., “[Ab-initio Determination of Light Hadron Masses](#),” *Science* **322**, 1224–1227 (2008), open manuscript. Consult the mass-spectrum results and discussion of continuum and physical-mass extrapolations. Quark masses and the overall physical scale are inputs; this does not assign all mass to a single geometric flux-tube component.

[4] Asymptotic freedom.

D. J. Gross, “[The Discovery of Asymptotic Freedom and the Emergence of QCD](#),” Nobel lecture (2004), especially “Implications of Asymptotic Freedom.” For the precise running-coupling equation, see source 2, §9.1.1, Eq. (9.3). A weak-coupling result at short distance is not by itself a long-distance confinement proof.

[5] Flux tubes, a static potential, and string breaking.

G. S. Bali et al. (SESAM), “[Observation of String Breaking in QCD](#),” *Physical Review D* **71**, 114513 (2005). The system has static quark sources and dynamical light quarks. The linear potential is an intermediate-regime model; the final states after breaking are color neutral. It is not a rigid-string model of a physical proton.

[6] What the unsolved mathematical problem actually asks.

A. Jaffe and E. Witten, “[Quantum Yang–Mills Theory](#),” §4 (printed p. 6) for existence and the spectral-gap statement; §5 distinguishes extensions including confinement. The full statement treats any compact simple gauge group on four-dimensional spacetime with specified axiomatic properties. The [Clay problem-status page](#) still lists the missing proof (checked 11 September 2026). This is not the claim that gluons acquire an ordinary particle mass.

[7] A scale-dependent-coupling analogue outside nuclei.

P. Coleman, “[Phil Anderson’s Magnetic Ideas in Science](#)” (2015), §3.5, printed p. 10. The Kondo example supplies the established mechanism of scale-dependent antiferromagnetic exchange and collective impurity-spin screening. Its use as a prompt for comparing nuclear scales is the author’s extrapolation.

[8] The QCD-to-nuclear-force bridge.

R. Machleidt and D. R. Entem, “[Chiral Effective Field Theory and Nuclear Forces](#),” *Physics Reports* **503**, 1–75 (2011). See the introductory discussion and sections on the chiral expansion, pion exchange, contact interactions, and many-nucleon forces. This is a systematic effective description with fitted inputs and quantified truncation errors.

Open sources, continued

[9] Nuclear notation and the binding-energy trend.

P. Cappellaro, [MIT 22.02, Chapter 1: Introduction to Nuclear Physics](#) (2012), §1.1.1 and §1.2. The liquid-drop explanation is qualitative here; no fitted mass formula is used to predict light-nucleus binding energies or the exact peak location.

[10] The deliberately short weak-interaction detour.

L. E. Marcucci, R. Schiavilla, and M. Viviani, “[The Proton–Proton Weak Capture in Chiral Effective Field Theory](#)” (2013). The solar reaction is not free proton decay, nor the only physical route by which deuterium can form.

[11] Numerical binding-energy inputs.

Atomic Mass Evaluation 2020, [mass_1.mas20](#) from the IAEA Atomic Mass Data Center. Rows $(Z, A) = (1, 2), (1, 3), (2, 4), (26, 56), (28, 62)$ provide the relevant values; the binding column is *keV per nucleon*, so multiply by A and divide by 1000 for total MeV. Atomic-mass conventions require electronic-binding corrections for high-precision bare-nucleus energies; they are immaterial to the 0.001 MeV inputs used here.

[12] Units and temperature conversion.

NIST, [SI Brochure, §2: The International System of Units](#). Exact defining values: $e = 1.602176634 \times 10^{-19}$ C and $k_B = 1.380649 \times 10^{-23}$ J K $^{-1}$. Their quotient gives $k_B = 8.617333262 \dots \times 10^{-5}$ eV K $^{-1}$. Together with NIST’s [CODATA table entry for vacuum electric permittivity](#), these give $e^2/(4\pi\epsilon_0) \simeq 1.440$ MeV fm.

[13] Two-body kinematics and the meaning of cross section.

Particle Data Group (2025), “[Kinematics](#),” §49.3 (invariant amplitudes), §49.4.2 (two-body final states), and §49.5 (cross sections). The main-text recoil result is a leading nonrelativistic estimate, not an exact relativistic energy partition.

[14] The quantum law used in the barrier illustration.

R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. III, Chapter 16, especially the Schrödinger-equation discussion. Our constant-barrier ODE illustrates penetration only; it is not used as a numerical D–T cross-section model.

[15] Amplitudes and interference.

Feynman, Leighton, and Sands, Vol. III, Chapter 3: [Probability Amplitudes](#), especially the rules for combining amplitudes. The main text uses a normalized wave-packet basis to avoid assigning a finite probability to one exactly specified continuum momentum.

[16] A real microscopic D–T calculation and its limits.

G. Hupin, S. Quaglioni, and P. Navrátil, “[Ab Initio Predictions for Polarized Deuterium–Tritium Thermonuclear Fusion](#),” *Nature Communications* **10**, 351 (2019). See Results, Fig. 1, and Methods: the “NCSMC-pheno” result includes a -5 keV resonance centroid correction. No polarized-fuel enhancement is assumed in our opening.

[17] Action and the sum over histories.

Feynman, Leighton, and Sands, Vol. II, Chapter 19: [The Principle of Least Action](#), concluding quantum discussion. This supplies an accessible conceptual entry, not a construction of the QCD functional measure.

[18] Later engineering connection.

J. P. Freidberg, F. J. Mangiarotti, and J. Minervini, “[Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?](#)” (2015), MIT report PSFC/JA-15-21. This is the subsequent design target, not the source of a confinement proof or a universal temperature threshold.

Sources for the speculative bridge

[19] The quantum liquid drop and induced nuclear interactions.

G. Potel Aguilar and R. A. Broglia, *The Nuclear Cooper Pair: Structure and Reactions* (Cambridge University Press, 2022), Chapter 1, printed pp. 1–4 for liquid-drop cohesion and surface energy, and pp. 8–17 for particle–vibration coupling and induced interactions. These supply the physical examples used in the proposed comparison across scales.

[20] BESIII’s 2026 glueball result.

BESIII Collaboration, [Lightest \$0^{-+}\$ Glueball as Dominant Constituent of \$X\(2370\)\$](#) (submitted 22 July 2026). The analysis combines mass, spin-parity, production, and decay evidence to identify a dominant pseudoscalar-glueball component in $X(2370)$. The result was announced at ICHEP on 5 August 2026, as reported by the [Institute of High Energy Physics](#).

[21] The historical proton–electron neutron conjecture.

J. Chadwick, [The Neutron and Its Properties](#), Nobel Lecture (12 December 1935), opening discussion of Rutherford’s 1920 conjecture. H. Yukawa, [Meson Theory in Its Developments](#), Nobel Lecture (12 December 1949), opening discussion of early attempts to explain nuclear binding through electron or electric-charge exchange by analogy with chemical bonding. These are the historical origins of the electromagnetic question posed here.

[22] A modern negative-cloud picture.

J. Friedrich and Th. Walcher, [A Coherent Interpretation of the Form Factors of the Nucleon in Terms of a Pion Cloud and Constituent Quarks](#), *European Physical Journal A* **17**, 607–623 (2003). This phenomenological account describes charge structure through constituent quarks and a pion cloud; the virtual proton–negative-pion component gives the modern image used in the comparison.